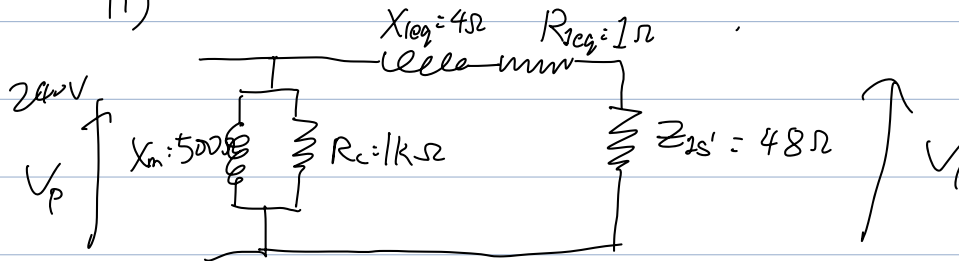


T₃

Q1. i) $TR = 240 : 12 = 20$

ii)



$$Z_{s'} = (TR)^2 \cdot Z_s = 20^2 \times 0.12 = 48 \Omega$$

$$Z_T \approx R_{seq} + jX_{seq} + Z_{s'}$$

$$= 49 + 4j \Omega$$

$$= 49.16 \angle 4.67^\circ \Omega$$

$$I_1 = \frac{V_p}{Z_T} = 4.88 \angle -4.67^\circ \text{ A}$$

$$I_s = (TR) \cdot I_1 = 97.6 \angle -4.67^\circ \text{ A}$$

$$\Rightarrow |I_s| = 97.6 \text{ A}$$

iii) $V_s = I_s \cdot R_s = 11.7 \angle -4.67^\circ \text{ V}$

$$\Rightarrow |V_s| = 11.7 \text{ V}$$

iv) Iron loss:

$$P_c = \frac{V_p^2}{R_c} = \frac{240^2}{1k} = 57.6 \text{ W}$$

Copper loss:

$$P_{cu} = I_1^2 R_{seq} = 4.88^2 \times 1 = 23.8 \text{ W}$$

$$P_{out} = I_s^2 R_s = 1143 \text{ W}$$

$$Q2. \quad VA = \frac{12 - 11.7}{12} \times 100\% = 2.5\%$$

$$\text{Efficiency} = \frac{1143}{1143 + 57.6 + 23.8} \times 100\% = 93.4\%$$

Max E occurs when

$$Q3. \quad P_c = P_{cu} \Rightarrow \frac{V_p^2}{R_c} = I_1^2 R_{1eq} \Rightarrow I_1 = \frac{V_p}{\sqrt{R_c R_{1eq}}} = 7.59 \text{ A}$$

$$I_1 = \frac{V_p}{R_{1eq} + jX_{1eq} + Z_{1s'}} \quad \text{Assume } Z_{1s'} \Rightarrow X_{1eq}$$

$$\Rightarrow Z_{1s'} \approx \frac{V_p}{I_1} - R_{1eq} = 30.62 \Omega$$

$$\Rightarrow Z_s = \frac{Z_{1s'}}{(TR)^2} = 0.0766 \Omega$$

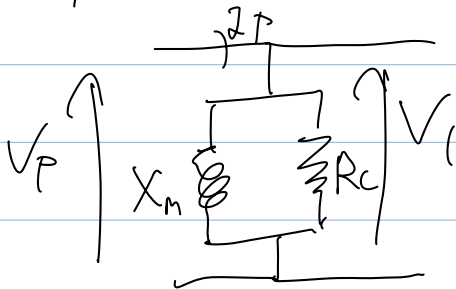
$$V_1 = I_1 \cdot Z_{1s'} = 232.4 \text{ V}$$

$$\text{Power: } I_1 V_1 = 1.764 \text{ W}$$

$$E = \frac{1.764}{1.764 + 57.6 \times 2} = 93.9\%$$

$$Q4. \quad TR = \frac{240}{110} = 2.18$$

open circuit:



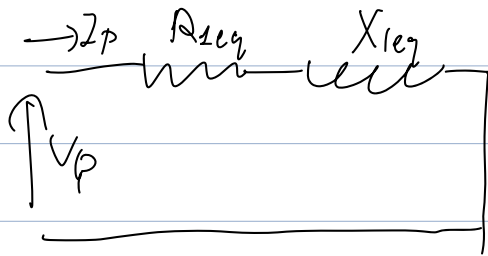
$$\frac{V_p^2}{R_c} = P_{in} \Rightarrow R_c = 576 \Omega$$

$$S = I_p V_p = 408 \text{ VA}$$

$$Q_p = \sqrt{S^2 - P^2} = 395 \text{ VAR}$$

$$\text{From } Q_p = \frac{V_p^2}{X_m} \Rightarrow X_m = 146 \Omega$$

Short circuit.



$$\text{From } P_{in} = I_p^2 R_{eq} \Rightarrow R_{eq} = 1.4 \Omega$$

$$S_p = I_p V_p = 378 \text{ VA}$$

$$Q_p = 364 \text{ VAR}$$

$$\text{From } Q_p = I_p^2 \cdot X_{eq} \Rightarrow X_{eq} = 3.15 \Omega$$

$$V_{rms} = \frac{\sqrt{2}}{2} N \omega B_{pk} A_c$$

$$V_{pk} = N \omega B_{pk} A_c$$

$$\Rightarrow V_{rms} = 4.44 N f B_{pk} A_c$$

$$V_{pk} \cos \omega t = N \omega B_{pk} \cos \omega t A_c$$

$$= 4.44 \times 200 \times 50 \times 1.3 \times A_c$$

$$= N \cdot \frac{d(B_{pk} \sin \omega t A_c)}{dt}$$